

Resolution Geometry: Reconstruction and Identifiability

The Geometry is Recovered from Observable Data up to Gauge,
and Fisher Information Alone is Not Enough

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Abstract

We settle the reconstruction question for resolution geometry: to what extent is the geometry recoverable from observable data? The answer has two parts, one positive and one cautionary, and both are exact. *Positive:* the resolution geometry $\mathcal{G} = (O; R, N, \Sigma)$ is a canonical function of the observable data (J, Σ, g) —the resolved sector is $\text{Im } J$, the null sector is the g -orthogonal complement $\ker J^*$, and the covariance is transported by push-forward—so it is uniquely determined, and equivariant under admissible (metric-preserving) changes of observational representation. It is therefore identifiable up to gauge, and an adversarial battery of five thousand random alternative decompositions, gauge perturbations, and independent numerical routes finds no non-identifiability witness beyond gauge. *Cautionary:* Fisher information alone does *not* identify the geometry. The Fisher metric is an inverse Schur complement of the covariance, a strictly coarser and many-to-one summary; when the resolved–null coupling vanishes it is provably blind to the null block, so infinitely many non-isomorphic geometries share one Fisher metric. Reconstruction therefore requires the full covariance, not the response metric. The result is modest in the positive direction—well-posedness and equivariance—and its weight is stated as such; the substantive content is the demonstrated insufficiency of Fisher data.

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1 The reconstruction question

Given what a measurement actually delivers—a forward-response Jacobian J and an observation covariance Σ on an observation space with metric g —is the resolution geometry it induces well defined, and can it be recovered unambiguously? Two failure modes must be ruled out: hidden *non-identifiability* (genuinely different geometries producing the same observable data), and

reliance on an *insufficient summary* (recovering the geometry from a compressed statistic that does not in fact determine it). We address both.

Non-claim 1.1 (Weight of the positive result). The positive reconstruction statement (Section 2) is a well-posedness and equivariance theorem; it is elementary. We state it because a foundational framework must establish that its central object is a genuine invariant of observable data rather than an artifact of representation, and because the substantive companion result—the insufficiency of Fisher data (Section 4)—is meaningful only once the positive statement is in place.

2 Reconstruction from observable data

Definition 2.1 (Observable data and induced geometry). The observable data are (J, Σ, g) : a Jacobian $J : T_\theta \Theta \rightarrow O$, a covariance $\Sigma \in \text{PSD}(O)$, and a fixed observation metric g on O . The induced resolution geometry is $\mathcal{G}(J, \Sigma) = (O; R, N, \Sigma)$ with

$$R = \text{Im}(J), \quad N = \ker(J^*) = R^{\perp_g},$$

where J^* is the g -adjoint. An *admissible gauge* is a g -isometry $Q \in \text{O}(g)$, acting by $J \mapsto QJ$, $\Sigma \mapsto Q\Sigma Q^*$.

Theorem 2.2 (Reconstruction and gauge-equivariance). *The assignment $(J, \Sigma, g) \mapsto \mathcal{G}(J, \Sigma)$ is well defined and canonical: every constituent of $\mathcal{G}$ is uniquely determined by the data. It is equivariant under admissible gauge, i.e. for $Q \in \text{O}(g)$,*

$$\mathcal{G}(QJ, Q\Sigma Q^*) = Q \cdot \mathcal{G}(J, \Sigma), \quad R' = Q(R), \quad N' = Q(N), \quad \Sigma' = Q\Sigma Q^*.$$

Hence the geometry is identifiable up to admissible change of observational representation.

Proof. $R = \text{Im}(J)$ is fixed by J . With g fixed the adjoint J^* is determined, so $N = \ker(J^*) = (\text{Im } J)^{\perp_g} = R^{\perp_g}$ is determined, and Σ is itself. For Q invertible, $\text{Im}(QJ) = Q(\text{Im } J)$, so $R' = Q(R)$; and $(QJ)^* = J^*Q^*$ with Q^* invertible gives $\ker((QJ)^*) = Q(\ker J^*)$, so $N' = Q(N)$. Covariance transports by push-forward, $\Sigma' = Q\Sigma Q^*$ (a $(2,0)$ -tensor). Thus each constituent transforms naturally under Q . $\square$

Remark 2.3 ($N = \ker J^*$ is the metric-orthogonal complement). The identity $(\text{Im } J)^{\perp_g} = \ker(J^*)$ holds because $J^*w = 0 \iff \langle Jv, w \rangle_g = 0 \forall v$. The null sector depends on g through the adjoint; this is the correct metric-explicit definition.

3 No non-identifiability beyond gauge: an adversarial battery

Theorem 2.2 says the geometry is determined; we confirm empirically that no hidden alternative decompositions exist. On a base system ($\dim O = 32$, $\dim \Theta = 10$, $\text{rank } J = 8$, $\dim R = 8$, $\dim N = 24$, $\Sigma \succ 0$, data coupling $\kappa_{RN} = 0.466$) we ran the following attacks.

Attack	Result	Verdict
Random alternative decompositions: 5000 candidate (R', N') with $\dim R' = r$	0 accidental valid; min. validity defect 1.55, min. projector distance 3.17	no witness
Observable gauge equivariance: 1000 random $Q \in O(g)$, transport back	0 failures; projector error $\leq 5 \times 10^{-15}$	confirms Thm 2.2
Independent reconstruction routes: SVD vs QR resolved projector	error 3.5×10^{-15}	route-independent

Table 1: Reconstruction battery. No non-identifiability witness beyond gauge. The uniqueness of $R = \text{Im } J$ is in fact forced: any r -dimensional R' with $\text{Im } J \subseteq R'$ equals $\text{Im } J$ since $\dim \text{Im } J = r$.

Proposition 3.1 (Rigidity). *Exact observable data (J, Σ, g) reconstruct $R = \text{Im } J$, $N = \ker J^*$, and the covariance block geometry $(\Sigma_R, \Sigma_N, C_{RN})$ rigidly; the only residual freedom is admissible gauge. Equivalently, two data sets induce isomorphic geometries iff they are related by a $Q \in O(g)$.*

Proof. Immediate from Theorem 2.2: the constituents are determined, and the orbit of the induced geometry under representation change is exactly the $O(g)$ -orbit. The battery exhibits no candidate outside this orbit. $\square$

4 Fisher information alone is insufficient

The substantive cautionary result: the response (Fisher) metric is not enough to reconstruct the geometry, so reconstruction genuinely requires the full covariance.

Theorem 4.1 (Fisher insufficiency). *For a Gaussian model the response metric is the inverse Schur complement pulled back by the Jacobian,*

$$\mathcal{I} = J^\top \Sigma^{-1} J = J_R^\top S_R^{-1} J_R, \quad S_R = \Sigma_R - C_{RN} \Sigma_N^{-1} C_{RN}^\top.$$

When $C_{RN} = 0$ this reduces to $J_R^\top \Sigma_R^{-1} J_R$, independent of the null block Σ_N . Hence the family of resolution geometries sharing a fixed (J_R, Σ_R) and $C_{RN} = 0$, with Σ_N ranging over all positive matrices, is a continuum of pairwise non-isomorphic geometries with identical Fisher information. The Fisher metric therefore does not identify the geometry.

Proof. The Gaussian Fisher information is $J^\top \Sigma^{-1} J$; block inversion gives $(\Sigma^{-1})_{RR} = S_R^{-1}$. At $C_{RN} = 0$, $S_R = \Sigma_R$, so $\mathcal{I}$ is independent of Σ_N ; varying Σ_N changes the geometry (different null block) while fixing $\mathcal{I}$. $\square$

Remark 4.2 (The general compression). Beyond the $C_{RN} = 0$ case, $\mathcal{I}$ compresses the $m(m+1)/2$ covariance parameters into the $\leq \binom{r+1}{2}$ of an $r \times r$ form; it is many-to-one whenever $q \geq 1$. A representative perturbation that inflates the null block and alters the coupling changes the full covariance by a large relative residual (≈ 1.65) while moving the Fisher metric by a smaller one (≈ 0.84): Fisher is strictly less sensitive than the full data, and blind in the $C_{RN} = 0$ limit.

Corollary 4.3 (Reconstruction requires the covariance). *The resolution geometry is reconstructable from (J, Σ, g) (Theorem 2.2) but not from $(J, \mathcal{I}, g)$ (Theorem 4.1). The full covariance is necessary; the response metric is not sufficient.*

5 Scope and non-claims

Scope. The results concern *exact* observable data. Reconstruction from *noisy or finite-sample* estimates of (J, Σ) is a separate statistical problem (estimator consistency, sample complexity) and is not treated here. The Fisher-insufficiency theorem is stated for the Gaussian model.

Non-claims. No claim is made that reconstruction is possible from partial observation, from the response metric, or from any summary coarser than the full covariance. The rigidity result is identifiability up to gauge, not a statement about recoverability under noise.

6 Conclusion

Reconstruction of a resolution geometry is well posed and rigid: the geometry is a canonical, gauge-equivariant function of exact observable data, identifiable up to admissible representation change, with no hidden non-identifiability. The one substantive caveat is that the response (Fisher) metric—the quantity one is most tempted to reconstruct from—is provably insufficient: it is an inverse Schur complement, blind to the null block when the coupling vanishes and many-to-one in general. Reconstruction is therefore a statement about the full covariance, and the response metric is a strictly weaker summary. This closes the identifiability question and fixes, precisely, which observable object the geometry actually depends on.

References

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